

Math 54 Discussion Worksheet

Name: _____

ID: _____

Problem 1

Suppose that A is invertible and orthogonally diagonalizable. Show that A^{-1} is also orthogonally diagonalizable.

Problem 2

Show this if A is a symmetric matrix, then so is A^2 . Also, show that if A is orthogonally diagonalizable, so is A^2 .

Problem 3

Let A and B be symmetric $n \times n$ matrices whose eigenvalues are all positive. Prove that all the eigenvalues of $A + B$ are positive.